



**Verified Carbon
Standard**

MONITORING REPORT *HYDROELECTRIC PROJECT EL EDÉN*

Project title	<i>Hydroelectric Project El Edén</i>
Project ID	1068
Monitoring period	<i>01-January-2021 to 31-July-2024</i>
Original date of issue	<i>07-November-2024</i>
Most recent date of issue	<i>10-April-2025</i>
Version	06
VCS Standard Version	4.7
Prepared by	ALLCOT AG

CONTENTS

PROJECT DETAILS	4
1.1 Summary Description of the Implementation Status of the Project	4
1.2 Audit History	5
1.3 Sectoral Scope and Project Type	5
1.4 Project Proponent	5
1.5 Other Entities Involved in the Project	6
1.6 Project Start Date	6
1.7 Project Crediting Period.....	6
1.8 Project Location	6
1.9 Title and Reference of Methodology.....	7
1.10 Double Counting and Participation under Other GHG Programs.....	7
1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	7
1.12 Sustainable Development Contributions	8
1.13 Commercially Sensitive Information.....	10
2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT.....	10
2.1 Stakeholder Engagement and Consultation	10
2.2 Risks to Stakeholders and the Environment	15
2.3 Respect for Human Rights and Equity.....	18
2.4 Ecosystem Health	21
3 IMPLEMENTATION STATUS	33
3.1 Implementation Status of the Project Activity	33
3.2 Deviations	34
3.3 Grouped Projects	36
3.4 Baseline Reassessment.....	36
4 DATA AND PARAMETERS	36
4.1 Data and Parameters Available at Validation	36
4.2 Data and Parameters Monitored.....	37
4.3 Monitoring Plan.....	39
5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	41
5.1 Baseline Emissions	41

5.2	Project Emissions	42
5.3	Leakage Emissions	43
5.4	GHG Emission Reductions and Carbon Dioxide Removals	43
APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION		46

PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Hydroelectric Project El Edén (hereafter, the project activity or project) is implemented by Proelétrica S.A.S E.S.P (hereafter, the project proponent) and has been verting electricity to the National Interconnected System (SIN, for its initials in Spanish) since the 16-February-2017.

The hydroelectric power plant utilizes water from the river “La Miel”, located in the river basin “La Miel” in the east of the Department of Caldas, Colombia (hereafter, the Host Country), about 125 kilometres from the City of Cali and close to the municipalities of Pensilvania, Marquetalia and Manzanares.

The project activity is a run-of-river hydroelectric plant that includes the construction and installation of a concrete weir and intake, a desander, channel, forebay/settling tank, high-pressure penstock, powerhouse and discharge channel. Two Pelton turbines with 10.319 MW and two synchronous generators with 11,154 kVA each. Each generator will be connected to a transformer to increase the voltage from 13.2 kV to 33 kV at the project site. The installed capacity of the plant at the interconnection is 19.9 MW, as per the document “Certificado de Cumplimiento de Código de Conexión”.

The most relevant implementation dates of the project are shown below:

- 28-June-2013: Signing of the Contract for the supply, assembly, and commissioning of electromechanical equipment.
- 4-November-2016: Request for registration in the Public Service Providers Registry (RUPS – for its translation in Spanish) to the Superintendence of Public Domiciliary Services.
- 22-November-2016: Approval of the request for registration in the RUPS.
- 28-February-2017: End of Commissioning Certificate.
- 28-February-2017: Certificate of compliance with the Connection Code (Resolution CREG 025 of 1995¹).

The net electricity generated by the project during the monitoring period 01-January-2021 to 31-July-2024 was 265,227 MWh which was delivered to the National Interconnected System (SIN),

¹ Access to Resolution CREG 025/1995:

https://gestornormativo.creg.gov.co/gestor/entorno/docs/resolucion_creg_0025_1995.htm

thus increasing the share of renewable energy and reducing a total of 81,476 tCO₂ during the same period.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	23-September-2013	VCS	ICONTEC	One year
Verification	16-February-2017–31-December-2020	VCS	KBS Certification Services Pvt. Ltd	Three years, ten months
Verification	01-January-2021–31-July-2024	VCS	AENOR	Three years, seven months

1.3 Sectoral Scope and Project Type

Sectoral scope ²	Energy (renewable/non-renewable)
Project activity type	Run-of-river hydro power plant
Sectoral scope	Not applicable since it is not an AFOLU project.
AFOLU project category ³	Not applicable since it is not an AFOLU project.
Project activity type	Not applicable since it is not an AFOLU project.

1.4 Project Proponent

The project proponent is Proeléctrica S.A.S E.S.P, which will be responsible for the construction, operation and maintenance of the project. Contact information is shown below.

Organization name	Proeléctrica S.A.S. E.S.P.
Contact person	Paula Andrea Villegas Londoño
Title	Legal Representative

² Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

³ See Appendix 1 of the VCS Standard

Address	Calle 100 No. 13-21 Piso 15, Edif. Mega Tower
Telephone	(+571) 7455060 - Ext. 570
Email	paula.villegas@genserpower.com

1.5 Other Entities Involved in the Project

Organization name	ALLCOT AG
Role in the project	Consultant
Contact person	Alexis Leroy
Title	CEO
Address	Steinhauserstrasse, CH-6300, Zug, Switzerland
Telephone	(+57) 322 209 8737
Email	ceo@allcot.com

1.6 Project Start Date

Project start date	16-February-2017
Justification	Date in which the project began to inject electricity to the SIN and therefore generating GHG emission reductions.

1.7 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	16-February-2017 to 15- February -2027

1.8 Project Location

The project is located in the Department of Caldas, Colombia. The project area includes the municipalities of Pensilvania, Marquetalia and Manzanares. The coordinates of the intake are E 885038, N 1079832 (datum Bogotá).

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	CDM: ACM0002	Consolidated baseline methodology for grid-connected electricity generation from renewable sources	13.0.0
Tool	CDM: TOOL01	Tool for the demonstration and assessment of additionality	7.0.0
Tool	CDM TOOL07	Tool to calculate the emission factor for an electricity system	3.0.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☒ Yes

☐ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes

☒ No

1.12 Sustainable Development Contributions

The Hydroelectric Project El Edén aims to replace fossil-fuel-based power generation with clean and renewable energy. The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action).

The oject contributes to the SDG 8.3. The maintenance and operation of the facilities necessitate outsourced services, leading to increased employment opportunities.

By displacing emission-intensive grid electricity through the generation of zero emission hydroelectricity, the project aligns with SDG 7 (Affordable and Clean Energy).

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1 Total electricity produced by non-conventional renewable sources	Implemented activities to increase	265,227 MWh were produced by renewable sources.	During the project during the crediting period of 10-years, the project aims to generate 1,045 GWh.
2)	8.5	8.5.1 Total number of jobs created	Implemented activities to increase	16 jobs were created with the project and all of them have been maintained during the monitoring period.	16 jobs were created with the project and It is expected to maintain all jobs throughout its lifespan.
3)	13.2	13.2.1 Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	81,476 tCO ₂ by displacing emission-intensive grid electricity through the generation of zero emission hydroelectricity.	The total expected emission reductions are 321,024 tCO ₂ during the 10-years crediting period.

1.13 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the monitoring report.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders have not changed since validation, and they were identified at verification.
Legal or customary tenure/access rights	There are no tenure or access rights to territories and resources that need to be considered in this case, as the project does not affect the territory of local communities or Indigenous Peoples. Additionally, no collective or conflicting rights related to the area have been identified.
Stakeholder diversity and changes over time	The project involves various stakeholder groups, including: Proelectrica: responsible for the operation of the hydroelectric plant. Local and national governments, responsible for regulating and overseeing the project. Local communities located in the project's area of influence. There have been no changes in stakeholder diversity.
Expected changes in well-being	No changes in well-being and other stakeholder characteristics were expected under the baseline scenario. In the absence of the El Edén Hydroelectric Project, ecosystem services in the La Miel River basin would have

	continued to deteriorate due to anthropogenic pressure and natural conditions in the area. • Soil regulation and erosion: Erosion would remain a serious issue, driven by the area's topography, deforestation, road construction without revegetation, and heavy rainfall. Landslides and soil degradation would continue to affect ecosystem stability. • Water availability and quality: The physical and chemical conditions of the water would progressively deteriorate due to uncontrolled water extractions and untreated wastewater discharges, impacting both biodiversity and human supply. • Vegetation cover and biodiversity: The transformation of forests into crops, mainly pasture and sugarcane, would increase the demand for wood for agricultural infrastructure. Only forests in inaccessible areas would be partially preserved, though still subject to selective logging. Generalist fauna species would persist, but endemic species could face local extinction. • Landscape and connectivity: Population growth and increasing resource demand would lead to further landscape degradation. The lack of adequate road infrastructure would keep some communities isolated, hindering their economic and social development.
Location of stakeholders	The local stakeholders of the El Edén Hydroelectric Project are located in various communities within the project's area of influence in the Caldas department, Colombia. Main municipalities: • Pensilvania • Manzanares • Marquetalia

	Community near the project site: • Bolivia (a district of Pensilvania)
Location of resources	The territories and resources used by the project are for the exclusive use of the proponent.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	The local stake holder consultation is part of the Environmental Impact Assessment (EIA) and its Environmental Management Plan (EMP), and Operation and Maintenance Plan (O&MP), according to the Colombian regulation. The project has started the local stakeholder consultation process in August 2010 and since then carried out a series of activities to inform and take into account the communities in the area of direct and indirect influence. A first stakeholder round was organized from August 2010 to May 2011 in the different communities. This process involved internal preparation, outreach events, and time for receiving feedback. The outreach events for this first round took place on October 29 and 30, 2010, in Pensilvania; on October 30, 2010, in Manzanares; and on November 4, 2010, in Marquetalia. A second round was carried out from June to August 2011. A third round with different meetings and visits during the period from March to May 2012 was carried out. Finally, a major event of the stakeholder process including the environmental impacts assessment, environmental management and modification of the environmental license was held on 14th of April 2013 at the John F. Kennedy School of the community of Bolivia which is close to the project location. The stakeholders were invited by letters of invitation to the presidents and members of the boards of communitarian actions that further informed the communities.
----------------------	--

Date(s) of stakeholder consultation	August 2010 to May 2011 June to August 2011 March to May 2012 14-April-2013
Communication of monitored results	El Edén has a procedure for the receipt and response of requests, complaints, claims, compliments, and suggestions (PQRFS). This procedure establishes a communication channel with different stakeholders, allowing for the timely receipt and response to various requests.
Consultation records	The Environmental Management Plan (EMP) included: -PMA-MS-01: Community and institutional information and participation: to evaluate the understanding of the information provided about the hydroelectric project and its characteristics during the different stages of the project. -PMA-MS-04: Information and sensibilization of the project to the local community: to inform the population that lives within the area of influence, about the activities of the project, during the studies, construction, and operation phases. -PMA-MS-06. Impact on third parties: to follow up on the requirements made by the populations within area of influence, about the Hydropower. Each one of these programs, have their own procedures / methodologies, indicators, goals and results which can be seen in detail in the Compliance Reports.
Stakeholder input	In order to demonstrate compliance of the Environmental License and its' environmental, operation and maintenance plans, Compliance Reports have been presented to Corpocaldas, the local environmental authority. Each report includes description of the project, maintenance activities, personnel responsible for the environmental activities, updated schedules and stages of the environmental and social plans and programs, as well as environmental compliances forms requested by Corpocaldas.

2.1.3 Free, Prior, and Informed Consent

Consent	No property rights' holders who are not project proponents are affected by the project and therefore the Free, Prior and Informed Consent wasn't needed within this project.
Outcome of FPIC	No property rights' holders who are not project proponents are affected by the project and therefore the Free, Prior and Informed Consent wasn't needed within this project.

2.1.4 Grievance Redress Procedure

The El Edén Hydroelectric Power Plant has received numerous communications through its PQRS (Petitions, Complaints, Claims, and Suggestions) system. Only two of these were formal complaints; the remaining ones were requests for collaboration from community members. Due to their high volume, these requests are not included in the report but are available in the Environmental Compliance Reports (ICAs).

Grievances received	Resolution and outcome
01/06/2021 A man comments that since the construction of the Central Station, a problem with the road elevation leaving the house below the level of the road, which is why the house floods when it rains.	A visit is made to the home and a request is made quote with construction contractor civilians.
06/08/2021 A woman comments that she has a water service of the Vereda of Bolivia but water has not been reaching his home for 5 months.	On July 26, 2021, a visit is made by officials of El Edén to verify the status of the aqueduct service in the Doña Nubiola's home. On 09/08/2021 the delivery of official letter CHE 2021-008 in which inform the lady that she does have water service but due to internal failures in its networks it is not working properly.

2.1.5 Public Comments

Summary of comments received	Actions taken
------------------------------	---------------

Extend to which the non qualified workers required for the construction will be from the region	All non qualified workers used for the construction will be locally hired.
Improvement of the roads in the area.	The management is assessing to make maintenance of other roads in the close area to the project.
Availability of water after implementation of the project and requirement to sign contracts for water security.	The environmental license is the security which is approved by COROCALDAS based on the detailed studies. Besides, it is clarified and explained that there is an environmental management plan that defines the required actions to assure minimization of the impacts

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The management team leading this project has extensive experience in the development of renewable energy projects, which includes both the construction and operation of similar facilities. Their deep understanding of applicable regulations, including environmental standards and public consultation requirements, positions them well for overseeing this initiative. The team's expertise ensures compliance with local and national laws, and their track record in conducting public consultations underscores their commitment to engaging with communities and stakeholders transparently.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Availability of water after implementation of the project.	The program for the management of ecological flow conservation is part of the environmental management plan. Its objective is to prevent any extraction or manipulation that exceeds the allowed limit for water capture, which could compromise the ecological integrity of the intervened system, leading to the loss of native species and services for society.

Risks to stakeholder participation	No risk identified	The Hydroelectric Project El Edén has social programs stated in the Environmental Management Plan (EMP), and Operation and Maintenance Plan (O&MP). Two of them guarantee all forms of participation. The program “PMA-MS-01. Community and institutional information and participation” evaluates the understanding of the information provided about the hydroelectric project and its characteristics during the different stages of the project. Furthermore, the program “PMA-MS-06. Impact on third parties” follows up on the requirements made by the populations within area of influence, about the Hydropower.
Working conditions	Occupational risks linked to the operation and maintenance tasks of the plant	The Hydroelectric Project El Edén has stated some internal policies to mitigate these risks such as: Internal work regulations; Action in case of blockage; Environment management guidelines for contractors; Work disconnection policy These policies and the area of talent’s management can guarantee the best working conditions.
Safety of women and girls	No risk identified	The monitoring and follow-up plan, as well as the environmental management plan, help minimize the risks to the safety of all nearby communities, including women and girls, that could be

		caused by the operation of the power plant.
Safety of minority and marginalized groups, including children	No risk identified	The monitoring and follow-up plan, as well as the environmental management plan, help minimize the risks to the safety of all nearby communities, including minority and marginalized groups, including children, that could be caused by the operation of the power plant. Additionally, the non-discrimination and child labor prevention policy ensure the necessary conditions to prevent any harm to children
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The monitoring and follow-up plan includes a program for the monitoring and follow-up of air resources and a project for the control of atmospheric emissions, air quality, and noise. It determines the concentrations of particulate matter, gases, and vapors through monitoring that complies with the specifications established by Colombian legislation. It also includes a program for the aquatic ecosystem resource, which involves monitoring the watercourse of tributaries in the section between water intake and discharge from the hydroelectric plant. Additionally, the environmental management plan addresses the integrated management of hazardous solid waste, the management and disposal of liquid waste, fuels, and oil, as well as the installation of individual

		septic tanks and a water treatment plant. Pesticides and fertilizers are not used in the project.
--	--	--

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁴	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.
Sexual harassment	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.
Gender equity in labor and work	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.
Forced labor	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.
Child labor	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.

⁴ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Human trafficking	No risk identified	The human rights and the non-discrimination and prevention of child labor policies, as well as the project's code of ethics, aim to reduce this risk.
-------------------	--------------------	---

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The Human Rights Policy of El Edén (PODE-05 POLITICA DE DERECHOS HUMANOS) affirms and develops its commitment to the Universal Declaration of Human Rights, ensuring careful adherence within the company. This policy complements the Code of Ethics and Conduct (PODE-09 CODIGO DE ETICA) and other corporate policies, where employees and executives are responsible for acting in accordance with the established guidelines. Additionally, the program PMA-MS-04 “Information and sensibilization of the project to the local community” informs the population that lives within the area of influence, about the activities of the project, during the studies, construction, and operation phases. Despite IPs in the project area are not identified, the company group have a guide to define the process of relationship with IPs. The document states that the group is committed to compliance with Human Rights, the International Charter of Human Rights, the conventions of the International Labor Organization (including the Convention 169), the principles of the Global Compact and as established in the Political Constitution and the international agreements ratified by Colombia (including the United Nations Declaration on the Rights of Indigenous Peoples), among which are respect for the ethnic communities is included

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	The project for information and training of personnel linked to the project, carried out during the construction phase, incorporates aspects related to the protection and management of archaeological heritage, considering that it was likely that archaeological artifacts or pre-Hispanic materials could be discovered during the land preparation phase, requiring safeguarding. In the event of finding ceramics, carved stones, stone engravings, bones, burials, or other elements that suggest belonging to pre-Hispanic communities that settled in the area, the worker or the person who locates the finding must: Leave the discovered material in its place. Immediately inform their supervisor or immediate superior. The supervisor will report to the field assistant. Apply the Archaeological Survey, Rescue, and Monitoring form. Additionally, during the operation phase, the information and training program for personnel linked to the project includes monitoring the incorporation of environmental protection tasks and individual environmental responsibility by workers to minimize the impact on historical and archaeological heritage.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The proof of title is given by the environmental license assigned to Central Hidroeléctrica El Edén S.A.S. E.S.P.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	The project does not impact property rights. This section is not applicable.
Benefit sharing during the monitoring period	The project does not impact property rights. This section is not applicable.

2.4 Ecosystem Health

Risk identified		Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	- Alteration of natural flora populations. - Interruption of biological corridors. - Habitat fragmentation. - Alteration of the ecological function of the forest.	The project includes the implementation of the following Monitoring and Follow-up Plans (PMS) and Environmental Management Plans (PMA) to mitigate the identified risks: PMA: Environmental Education (PMA_MB_03_P1): The objective is to contribute to the protection and environmental conservation and reduce the negative direct interaction with wildlife and its habitat (hunting, alteration of terrestrial habitats, pollution, etc.) in the La Miel River basin through talks and distribution of informational pamphlets to the local population. The following measures are established: Conduct scheduled talks to raise awareness among the population directly involved with the project, particularly during their stay in the temporary camps. In these meetings, workers and other people involved in the

		construction will be informed about the restrictions on hunting wildlife, the importance of wildlife in the ecosystem and human activities, the differences between venomous and non-venomous snake species, and logical explanations for beliefs (myths) related to certain animal species and why they are pursued. • Create illustrative pamphlets that guide actions aimed at conserving nature: waste disposal, conservation of riverbank buffers, responsible use of wildlife, and use of flora. These pamphlets will be distributed free of charge to students in schools and to the general community, accompanied by an awareness-raising talk. PMS: Program 1: monitoring of flora compensation and restoration • Project 1: Protective Reforestation: The objective is to assess the progress of protective reforestation established as compensation for the El Edén project. is project is 100% compliant, as it was carried out during the first years after construction. Currently, the company continues to develop associated actions, such as the setup and management of two (2)
--	--	---

		nurseries at the camp office and the intake area, rescuing the regeneration of forest species. Several of these specimens are donated to various local organizations. Among the recent activities, the initiative of the community from the La Soledad village stands out. This group, organized within the village aqueduct, has worked to protect the supplying microcatchment by planting six hundred (600) forest individuals, driven by concerns over the decreased flow due to the impacts of the El Niño phenomenon. This effort contributes to the ecosystem connectivity of the tributaries to the La Miel River. Establishment of Bamboo Groves: The objective is to evaluate the progress of bamboo groves established as a compensatory measure for the El Edén project. The project has 100% compliance given that it was carried out during the first years of construction. Program 2: habitat protection and conservation Linear Park (PMS_MB_02): The objective is to assess and maintain the linear park of El Edén. Within the framework of the Project,
--	--	--

		routine maintenance actions (spraying, mechanical and manual, and spraying) are carried out on the ecological trail in the catchment area.
Soil degradation and soil erosion	• Impact on soil stability and current land use • Alteration of the landscape	The project includes the implementation of the following measures considered in the Environmental Management Plan (PMA) to mitigate the identified risks: Program 1: monitoring and follow-up of soil resources • Control of Geotechnical Stability and Slopes Management Due to Erosive Processes: The objective is to monitor the geotechnical stability and slopes affected by erosion in the different construction fronts of the project to assess the effectiveness of the management measures. • This program was implemented after the construction phase. The clearing and stripping were carried out only in the areas strictly necessary for the adaptation of access routes and the construction of both preliminary and final works. The cleared material was placed in adjacent areas for decomposition and reintegration into the soil as organic matter.

		• The removal of the organic soil layer in excavation sites was conducted by extracting the topsoil along with the vegetation layer at the time of stripping. The vegetation layer was separated from the soil and placed in nearby areas for decomposition, while the organic soil was temporarily stored for restoration activities. These materials will be stored with full coverage to prevent the effects of water and wind. • Earthworks were carried out in a way that balanced cut and fill volumes, minimizing excavation surpluses as much as possible.. • Slope Management: The objective of the project is to define the actions to be taken for slope management in the areas that will be affected by the construction works, establishing a plan for the prevention and mitigation of instability effects. This project was completed during the construction phase.. To maintain the slopes that require stabilization, the following activities are recommended: • Maintenance and cleaning of the drainage systems that will be installed on the slopes, in order to avoid
--	--	--

		their deterioration and malfunction. • Profiling of the slopes to prevent the activation of erosive processes or mass wasting. • If cracks are found on the surfaces of the slopes, sealing them to prevent the infiltration of surface water or runoff. • Mulching of slopes to help control stability and prevent erosion that could be caused by water action; as well as to contribute to improvements in landscape development. • Revegetation and reshaping of the intervened areas. • Native species will be used in the revegetation whenever possible.
Water consumption and stress	• Impact on water quality due to increased particulate matter and contaminants • Hydrological changes in the section between intake and discharge • Impact on microbiological resources • Alteration of the water resource due to channel diversion, vegetation removal, excavations, domestic installations, and machinery maintenance	The project includes the implementation of the following Monitoring and Follow-up Plan (PMS) and Environmental Management Plans (PMA) to mitigate the identified risks: PMS: Monitoring of the tributary watercourse in the section between the intake and discharge of the hydroelectric plant (PMS_MF_05_P1): The project aims to define actions for the conservation of the La Miel River tributaries within the direct influence area of the hydroelectric

		plant. It also seeks to prevent alterations in the tributary watercourses and the reduction of their flow contribution to the main channel. Activities include: • Conducting flow measurements in the streams located within the Direct Influence Area. • Reviewing piezometric levels in the inspection wells of the tunnel's Direct Influence Area. • Performing a comparative analysis with the data recorded in the baseline study of the hydroelectric plant's Environmental Impact Assessment. PMA: Program 3: monitoring and follow-up of water resources • Project 1: Water Quality Control: The objective is to assess the physicochemical and hydrobiological conditions of the La Miel River at the sampling points defined in the water resource characterization, as well as in the watercourses potentially affected by the construction of access roads, main and secondary works, and the operation of camps, workshops, and facilities required for the construction and operation. The activities
--	--	--

		carried out follow the IDEAM Guidelines for water monitoring and tracking in all sampling and analysis processes: Guide for Water Monitoring and Tracking and Guide for Monitoring Discharges and Surface Waters. Additionally, at each selected sampling site, in situ parameters are measured (water temperature, dissolved oxygen, conductivity, and pH), and a sample is collected for analysis at a laboratory accredited by IDEAM. Once the results are obtained, the concentrations found in the water bodies are compared against current regulations.
	•	

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	The Environmental Management Plan establishes the Program 4: management of endangered and/or endemic species in the direct influence area, which includes Project 1: "Monitoring of the Populations of the Frog <i>Rheobates palmatus</i>, the Salamander <i>Bolitoglossa ramosi</i>, and the Lizard <i>Anolis tolimensis</i>, endemic species to Colombia, to Prevent and Mitigate the Project's Impact on them." The objective is to implement the environmental management of endangered species through monitoring to determine population sizes, habitat use, and specific requirements of these three endemic species present in the Direct Influence Area of the project, collecting information to prevent and mitigate the impacts of the works on their populations and ensure their permanence in the area.
--------------------	---

To prevent impacts, the project proposes:

- Conducting periodic censuses (during rainy and dry seasons) in the intervention area during the pre-construction phase (Study stage) to determine population size, habitat use, and the specific requirements of the species.
- Assessing whether habitat intervention is appropriate or not for the species and/or determining the management that should be applied based on presence data, population size, habitat use, and specific requirements. Considering the need to acquire land for compensation purposes.
- Conducting periodic monitoring (during Construction and Operation) to evaluate the effects of habitat fragmentation on the species.

The monitoring results have been favorable. For example, the latest monitoring revealed a rich avian diversity in the area of influence of the El Edén Hydroelectric Plant (CHEE). This diversity highlights the importance of this location as a vital habitat for birdlife.

Regarding mammalian fauna, four additional species were recorded during the monitoring. This is considered significant as it increases the available information on previously recorded species and helps assess the abundance and richness of the study area for conservation and restoration purposes.

As for the amphibian population, a new species, *Rhinella humboldti*, was identified. This species is endemic to Colombia. Additionally, the presence of other endemic species previously found in the monitoring, such as *Rheobates palmatus* and *Leucostethus fraterdanieli*, was confirmed.

Finally, concerning the reptile population, the results were not entirely conclusive. Over time, a total of 31 reptile species have been reported in the project. In the latest monitoring, 7 species and 9 individuals were recorded, along with the identification of a new species. High secondary vegetation was also identified. This type of

	vegetation is important for reptile diversity as it provides a stable and resource-rich environment for these species. However, no reptiles were reported in this type of coverage, which may be due to various factors, including climatic, methodological, and seasonal variables, among others.
Areas needed for habitat connectivity	The Environmental Management Plan establishes the Program 7 “Aquatic ecosystem resource”, which includes Project 1 “Management of Ecological Corridors (Longitudinal Connectivity)”. Its objective is managing the riverine ecological corridor, understood as the pathway that allows genetic flow through species migration, and to establish a plan for the prevention, mitigation, and compensation of impacts.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	The alteration of the habitat used by these species, as well as the reduction of available resources, can lead to a decline in the populations of <i>Rheobates palmatus</i>, <i>Bolitoglossa ramosi</i>, and/or <i>Anolis tolimensis</i>, or a possible local extinction of these species.	The project includes the implementation of the following Monitoring and Follow-up Plan (PMS) to mitigate the identified risks: The Program 3 “Monitoring of endangered and/or endemic species in the direct influence area” includes Project 1 on “Monitoring of the Populations of the Frog <i>Rheobates palmatus</i>, the Salamander <i>Bolitoglossa ramosi</i>, and the Lizard <i>Anolis tolimensis</i>, species endemic to Colombia, to prevent and mitigate the project's impact on them. The project involves identifying the potential habitats used by these species and their specific requirements, considering vegetation cover and the presence of water bodies. Using standardized

		methodology for herpetofauna inventories, the presence of these species in the area will be confirmed. Once the populations are identified, periodic censuses will be conducted to estimate population sizes and resource use. Subsequently, monitoring will continue regularly to implement measures aimed at preventing and mitigating population decline or potential local extinction of these species. • Visual censuses will be conducted every two years during the operation phase to assess the number of groups present and observe their behavior. • Twice a year, periodic censuses will be conducted, and population coverage percentages will be verified during the pre-construction phase (Study stage) to estimate the population size and habitat use of the species <i>Habia cristata</i> in forest patches that will be intervened or are near intervention sites (for road construction, etc.). • Twice a year, the population percentage in the study areas will be evaluated until construction begins. • Twice a year, species coverage in the study area will be assessed until construction begins.
--	--	---

		- Meetings with the local population will be held twice a year during construction to verify sightings of the species in the region. - Every three months, for the duration of the project, a local environmental promoter will survey the entire trail system and assess the population status through visual censuses, determining presence or absence of species.
Areas for habitat connectivity	Alteration of ecosystem connectivity due to the reduction of flow in the affected section.	The project includes the implementation of the following Monitoring and Follow-up Plan (PMS) to mitigate the identified risks: The Program 4 “Monitoring and follow-up of the aquatic ecosystem” includes Project 1 “Monitoring the Conservation of Ecological Flow”. Its objective is to establish actions to maintain an ecological flow that preserves the useful habitat and aquatic corridor for the fauna and flora present in the La Miel River between the intake and discharge sections.

2.4.2 Introduction of species

No adverse effects from the introduction of species or invasive species have been identified at validation.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Not applicable	Not applicable	Not applicable	Not applicable

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
Not applicable	Not applicable

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	Not applicable	Not applicable

2.4.3 Ecosystem conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	The Hydroelectric Project El Edén is not an ARR, ALM, WRC or ACoGS project. This section is not applicable.	The Hydroelectric Project El Edén is not an ARR, ALM, WRC or ACoGS project. This section is not applicable.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The Hydroelectric Project El Edén has been operating since 16-February-2017, and no mayor breakdowns have happened. The project continues to operate under normal conditions.

However, some observations of the daily reports show turbidity and /or contamination on the incoming water flow. Sometimes, even obstructing the pipe for measuring the flow.

There have been some short stops during the operation mainly due to electrical storms, low level of the river or mechanical issues. Most of these stops have been quickly solved and operation restored within a few minutes. Other partial but longer stops of the equipment have taken place for routine maintenance and have usually lasted for a few hours.

The most significant adverse event occurred on May 15, 2021, when a flood affected power generation for approximately 10 days. The operations and maintenance team promptly addressed the situation. Records indicate that at 18:00, the generating units tripped due to an electrical storm, activating the protection relays for the transformers and generators. This triggered the SCADA emergency shutdown sequence, alarms were reset, and by 18:18, power generation was restored. However, at 19:42, another trip occurred due to a second electrical storm, with the same protection relays responding. Additionally, various circuits in the area were disrupted but were later restored.

Following the event, maintenance work was carried out, including cleaning the intake grates of the weir, desilting the settling tank, and servicing the ecological flow gate. A detailed report, "CHEE_Informe avenida torrencial_V0 Mayo 2021," was prepared to document the incident.

Other minor maintenance events have been recorded in the daily O&M files which are shared as evidence.

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviations have been necessary for this Monitoring Report; therefore, this section does not apply.

3.2.2 Project Description Deviations

The main project description deviation for Hydroelectric El Edén is on the starting operation date which according to the VCS PD was supposed to be on the 01-January-2015 and ended up being on the 16 February-2017.

The expected construction date according to the VCS PD was June 2013 and the actual signing of the Contract for the supply, assembly, and commissioning of electromechanical equipment happened on the 28-June-2013, therefore, the construction started just as planned in the VCS PD. However, there were some difficulties during the construction phase, mainly on the excavation and underground activities, causing a delay on the starting operation date. The End of Commissioning Certificate was signed on the 28-February-2017. However, according to the SIN administrator, XM, the project Hydroelectric El Edén has been verting electricity to the SIN, on the 16-February-2017.

Thus, there is also a deviation on the project crediting period which was meant to start on the 01-09-2015. However, since the project only started generated GHG emissions reductions on the 16 February-2017, the project crediting period is 10 years, from 16-February-2017, date in which the project began to inject electricity to the SIN and therefore generating GHG emission reductions, to 15-February-2027.

There were also, some minor changes in the main equipment capacity (turbines, generators and transformers). However, it was stated in the VCS PD that the values proposed then, were based on the designs, and could vary slightly at real conditions. The capacity of the actual installed equipment is shown below:

- Turbines: 10,319 kW
- Generators: 11,154 kVA
- Transformers: 12,000 kVA
- Power nominal factor: 0.9

According to the installed capacity of the generators and their power factor, the installed capacity of the hydroelectric plant is 20.078 MW, which differs lightly from the installed capacity declared in the PDD.

These modifications are not considered significant, as they do not affect either the baseline or additionality. In the latter case, a sensitivity analysis was included in the registered PDD, raising the projected output from 104.5 GWh to 114.95 GWh, which would require increasing the capacity from 20.11 MW to 22.12 MW. These values are higher than the current installed capacity.

The sensitivity analysis was carried out by varying the following key parameters to analyze the impact on the equity

- o IRR: Energy generation (MWh): The quantity of energy generated is varied.
- o Investment costs (USD): the complete investment cost is varied.
- o O&M costs (USD/year): the complete O&M cost is varied.

The energy tariff is not varied since it will be fixed under the energy sales agreement.

The following Table 7 shows a sensitive analysis for the resulting IRR by varying the main parameters +/- 10%.

Variation of electricity generation	Base	+10.00
IRR	12.22%	13.21%

Variation of electricity tariff	Base	+10.00%
IRR	12.22%	13.32%

Variation of O&M costs	Base	-10.00%
IRR	12.22%	12.32%

As can be seen, the benchmark of 14.90% was not reached in these scenarios.

It was also stated in the VCS PD that the installed capacity of the plant at the interconnection point will be around 19.5 MW. According to the Commercial operation declaration, the installed capacity at the interconnection point is 19.9 MW.

Finally, the Project Proponent (PP) was modified in 2023. By signing a Deed of Partial Release dated September 1, 2023, the former PP CENTRAL HIDROELÉCTRICA EL EDÉN S.A.S E.S.P. was formally released from the representations, warranties, obligations, and liabilities assumed under the VCS Registration Deed of Representation, originally executed on September 20, 2013. In this regard, it is hereby informed that the currently recognized PP in Verra Registry is PROELECTRICA S.A.S E.S.P. This change has been duly formalized and recorded with Verra, ensuring the continuity of the project and compliance with the requirements established under the VCS program.

3.3 Grouped Projects

This is non-Grouped project; therefore, this section does not apply.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Ex ante combined margin CO ₂ emission factor for grid connected power generation in year y calculated using version 3.0.0 of the “Tool to calculate the emission factor for an electricity system”
Source of data	Calculated based on information provided by the grid operator XM.
Value applied	0.3072

Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” (version 3.0.0). The grid emission factor (EF_{grid, CM, y}) was calculated by the PP using the Tool to calculate the emission factor for an electricity system as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) factors, according to the following steps: Step 1. Identify the relevant electricity systems: For Hydroelectric Project El Edén case, the PP used the Colombian transmission system as the project electricity system, which is operated and maintained by XM. Step 2. Choose whether to include off-grid power plants in the project electricity system (optional). In step 2, the PP chose option I, only grid power plants are included in the calculation. Step 3. Select a method to determine the operating margin (OM). In the step 3, simple adjusted OM was chosen by the PP to calculate the operating margin emission factor, using ex-ante data vintage taking into account that in the Colombian electric system, the low-cost/must-run resources constitute more than 50% of total grid generation, hence the Simple OM method cannot be used. Step 4. Calculate the operating margin emission factor according to the selected method. Calculations of OM emission factor (step 4) were made as illustrated in the PD version 04, section 3.1, which is according to the tool specifications. Step 5. Calculate the build margin (BM) emission factor. In order to calculate the BM emission factor (step 5) option 1 (ex-ante) for the first crediting period was adopted, the build margin emission factor is calculated based on the most recent information available on units already built for sample group m at the time of PDD submission to the DOE for validation. The PP took the information from the latest official XM statistics. Step 6. Calculate the combined margin (CM) emission factor. Finally, combined margin was correctly calculated by weighted average method.
Purpose of data	Calculation of baseline emissions
Comments	--

4.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y

Source of data	Electricity meters														
Description of measurement methods and procedures to be applied	The meters installed are bi-directional, taking into account both the quantity of electricity supplied by the project plant to the grid and the quantity of electricity delivered to the project plant from the grid. In this case the individual values are not required. Monitoring procedures are implemented onsite or remote, using tele-metering technology. The operational team is in charge of taking the measurements and reporting to XM. The measurements were taken during the monitoring period: 01-January-2021—31-July-2024														
Frequency of monitoring/recording	Continuous measurement and at least monthly recording. Typically, the measured data is read once every 24 hours using tele-metetering technology (remotely) and reported to XM (grid operator and administrator).														
Value monitored	<table><tr><th>Year</th><th>EG_{facility,y} (MWh)</th></tr><tr><td>2021</td><td>86,661.50</td></tr><tr><td>2022</td><td>92,259.73</td></tr><tr><td>2023</td><td>53,153.45</td></tr><tr><td>2024</td><td>33,152.63</td></tr></table>			Year	EG _{facility,y} (MWh)	2021	86,661.50	2022	92,259.73	2023	53,153.45	2024	33,152.63		
Year	EG _{facility,y} (MWh)														
2021	86,661.50														
2022	92,259.73														
2023	53,153.45														
2024	33,152.63														
Monitoring equipment	A main and a backup meter were installed, both bi-directional and with the following characteristics for both meters: - Brand: Schneider Electric - Model: Power Logic ION8650 socket meter (2011) - Accuracy class: 0.2S - Digital fault recording: Simultaneous capture of voltage and current channels for sub-cycle disturbance transients. - Main meter serial number: MW-1509A351-02 - Backup meter serial number: MW-1509A352-02														
QA/QC procedures to be applied	Calibration was done according to the internal “Procedure for maintenance of power meters and transformers” and the VCS PD. <table><tr><th>Description</th><th>Main Meter</th><th>Back-Up Meter</th></tr><tr><td>Serial Number</td><td>MW-1509A351-02</td><td>MW-1509A352-02</td></tr><tr><td>Calibration date</td><td>15-September-2018</td><td>15-September-2018</td></tr><tr><td>Valid till</td><td>14-September-2021</td><td>14-September-2021</td></tr></table>			Description	Main Meter	Back-Up Meter	Serial Number	MW-1509A351-02	MW-1509A352-02	Calibration date	15-September-2018	15-September-2018	Valid till	14-September-2021	14-September-2021
Description	Main Meter	Back-Up Meter													
Serial Number	MW-1509A351-02	MW-1509A352-02													
Calibration date	15-September-2018	15-September-2018													
Valid till	14-September-2021	14-September-2021													

	Calibration date	29-October-2022	29-October-2022
	Valid till	28-October-2025	28-October-2025
	The calibrations were held by laboratory Veritest Ltda. accredited by ONAC under accreditation code 10-LAC-032. The frequency of calibration is 3 years according to Monitoring Plan stated in the VCS PD . The electricity generated during the period between 15-September-2021 and 29-October-2022 was adjusted with meter's error (refers to green highlighted cells in the sheet "El Edén Generation Summary").		
Purpose of the data	Calculation of baseline emissions		
Calculation method	Not applicable		
Comments	All data collected as part of the monitoring process is archived electronically and kept at least for two years after the end of the last crediting period.		

4.3 Monitoring Plan

The Monitoring Plan basically consists in the procedures to meter the project's electricity generation. Since this is the only parameter to be measured and the emission factor is already set ex ante, the process is pretty straightforward, as described in the following.

Electricity generation of the project activity

Monitoring procedures are implemented onsite or remote, using tele-metering technology. The operational team is in charge of taking the measurements and reporting to XM. A main and backup meters are installed at the interconnection point of the project with the SIN. The energy meters in the substation are read via dedicated software every 24 hours. A daily report is then generated and sent to XM.

As indicated, there is a main electricity meter and a backup meter, which assures correct measurement in case of failures of the main electricity meters. If any other emergency would not allow measuring correctly the power generation of the plants or if data would be missing as a result of a failure in the monitoring process, no emission reductions shall be claimed during that period until guaranteeing again correct function of the meters and having reliable data.

The information is backed up by the operational team frequently. The data is included in an excel spreadsheet for emission reductions calculations on a monthly basis. All data collected as part of the monitoring process are archived electronically and kept at least for two years after the end of the last crediting period.

The following scheme shows the power plants, the substation and the metering points:

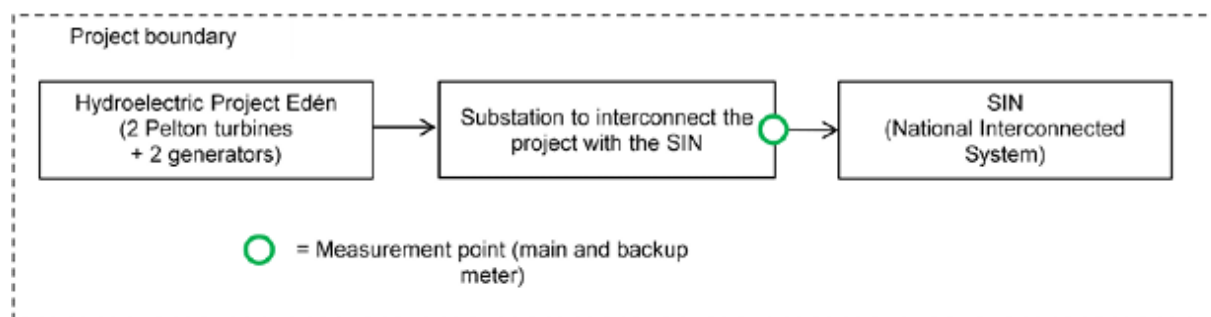


Figure 1. Simplified scheme of the monitoring.

QA/QC measures

The meters are calibrated maximum every three years according to Monitoring Plan as per it is indicated in the VCS PD. Calibration tasks follow national standards and are in accordance with the calibration instructive specified in Colombian standard NTC 4,856 for electricity metering devices⁵. In case both meters fail, no emission reductions will be claimed during that period until having again data from the main or backup meter.

Personnel responsible for monitoring

Figure 2 shows the organizational chart of the monitoring.

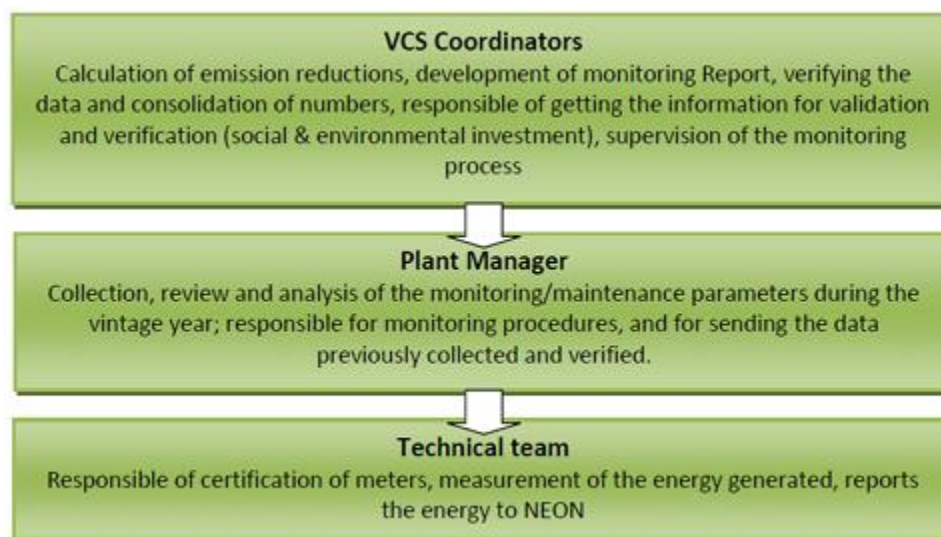


Figure 4. Operational structure of the monitoring plan

Responsible personnel:

⁵ Colombian standard NTC 4856: <https://tienda.icontec.org/gp-verificacion-inicial-y-posterior-de-medidores-de-energia-electrica-ntc4856-2018.html>

- The VCS Coordinators supervise the monitoring process and are in charge of compiling the monitoring data in an excel spreadsheet and calculating the emission reductions of the monitoring period. They also develop the monitoring report in accordance with the VCS rules.
- The Plant Manager is responsible for verification of energy measurement. He is responsible for checking and verifying the meter readings.
- The Instrumentation and Control Engineers are responsible for electricity generation reading and for processing the energy produced by the project from the meters installed at the substation. Records of the meter are downloaded in a spreadsheet for measurement control and the data discharged from the meter is stored electronically and sent to XM (<https://sinergox.xm.com.co/Paginas/Home.aspx>).

Personnel who carry out monitoring tasks are familiar with the basic monitoring requirements and structures. New personnel have to participate in a basic training in order to get familiarized with the monitoring procedures. Since the main monitoring tasks, i.e. the measurement of the energy production, the calibration of energy meters, and the reporting of the energy generation, are carried out independently from the VCS as part of the daily operation, no specific training is required. The VCS Coordinators are supported by an external consultant if necessary, in order to assure correct application of the monitoring procedures. They also carry out corrective actions if any inconsistency is identified and train the Plant Manager and the Technical team, if necessary.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The baseline scenario and the emission rate calculation are based on the electricity that otherwise would have been generated by the plants connected to the grid and by addition of future plants. According to the methodology, no leakage emissions are considered.

Baseline emissions are calculated by multiplying the combined margin emission factor ($EF_{grid,y}$, in tCO₂e/MWh) by the electricity generated by the proposed project activity during the year y (EG_y , in MWh).

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation by the project activity would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The detailed calculations and data of the baseline emissions are presented in the Excel file “El Edén ER MP2_V3”, for which the following equation (11 as per ACM0002) are presented.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using version 3.0.0 of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Since it is a green-field project, according to Eq. (12) of ACM0002,

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The expression $EG_{PJ,y}$ was used in the VCS PD and it is also used in this Monitoring Report. This parameter was measured during the monitoring period as established in the VCS PD. For more detail, please refer to the file: “El Edén ER MP2_V3”

The combined margin emission factor ($EF_{grid,CM,y}$) was calculated according to the guidance in the “Tool to calculate the emission factor for an electricity system” (version 3.0.0) in the VCS PD and set as a fix parameter for this MR. Hence the value applied was 0.3072 tCO₂/MWh.

The baseline emissions are calculated as shown in the following table:

From	To	Ex Post $EG_{facility,y}$	Ex Ante $EF_{grid,CM,y}$	BE_y
		MWh/yr	tCO ₂ /MWh	tCO ₂
1/1/2021	31/12/2021	86,661.50	0.3072	26.622
1/1/2022	31/12/2022	92,259.73	0.3072	28.342
1/1/2023	31/12/2023	53,153.45	0.3072	16.328
1/1/2024	31/7/2024	33,152.63	0.3072	10.184
1/1/2021	31/7/2024	265,227.31	0.3072	81.476

5.2 Project Emissions

As per the applied ACM0002 version 13.0.0, for hydro power project activities that result in new reservoirs, project proponents do not need to account for CH₄ and CO₂ emissions from the

reservoir, if the power density of the project activity (PD) is greater than 10 W/m² according to Equation (5) of the mentioned methodology.

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m²).

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero.

Since the project activity has no reservoir, this step is not required, and the project emissions are zero: PE_y = 0

5.3 Leakage Emissions

No leakage emissions are considered as provided in the methodology:

$$LE_y = 0$$

5.4 GHG Emission Reductions and Carbon Dioxide Removals

According to ACM0002 v.13.0.0 (eq. 11), emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂e/yr).

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
----------------	---	--	--	-------------------------------------	-----------------------------------	---------------------------------

01-Jan-2021 to 31-Dec-2021	26,622	0	0	26,622	0	26,622
01-Jan-2022 to 31-Dec-2022	28,342	0	0	28,342	0	28,342
01-Jan-2023 to 31-Dec-2023	16,328	0	0	16,328	0	16,328
01-Jan-2024 to 31-Jul-2024	10,184	0	0	10,184	0	10,184
Total	81,476	0	0	81,476	0	81,476

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-Jan-2021 to 31-Dec-2021	32,102	26,622	-17.07%	Lower-than-expected electricity generation due to lower-than-anticipated hydrological resources and the event mentioned in Section 3, which took the plant out of operation for 10 days.
01-Jan-2022 to 31-Dec-2022	32,102	28,342	-11.71%	Lower-than-expected electricity generation due to lower-than-anticipated hydrological resources.
01-Jan-2023 to 31-Dec-2023	32,102	16,328	-49.14%	Lower-than-expected electricity generation due to the El Niño phenomenon, which significantly reduced water availability.
01-Jan-2024 to 31-Jul-2024	18,645(*)	10,184	-45.38%	Lower-than-expected electricity generation due to the El Niño phenomenon, which significantly reduced water availability.
Total	114,951	81,476	-29.12%	

(*) It was calculated as follows:

Amount of days between 01-Jan-2024 to 31-Jul-2024: 213- 58% of calendar year.

Ex-ante estimated reductions: $32,102 \text{ tCO}_2\text{e} * 58\% = 18,645 \text{ tCO}_2\text{e}$

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

No commercially sensitive information has been excluded from the public version of the monitoring report.

Section	Information	Justification
Not applicable	Not applicable	Not applicable